

DevOps Production Interview Questions

Merged & Deduplicated Master Set — Jenkins, Terraform, Docker, Kubernetes, AWS EKS, Ansible + Real Disaster/Incident Scenarios

Note: Dono source PDFs compare karke jo questions concept-wise duplicate the (jaise 'etcd corrupt recovery', 'state file lost', 'OOM debugging', 'terraform destroy prevention', 'secrets in image layers', etc.) unhe generic Part A se hata diya gaya hai — unka zyada detailed, scenario-based version Part B me already maujood hai. Isse koi bhi question do baar nahi milega.

Contents

PART A — Concept-Level Production Questions

- Jenkins (50 Questions)
- Terraform (37 Questions)
- Docker (44 Questions)
- Kubernetes (43 Questions)
- AWS EKS (50 Questions)
- Ansible (30 Questions)

PART B — Real Disaster / Incident Scenario Questions

- Kubernetes — Cluster & Workload Failures (30 Questions)
- Kubernetes — Secrets & Access Management (25 Questions)
- Terraform — State File & Infrastructure Disasters (30 Questions)
- Database Disaster Recovery (25 Questions)
- GitOps, ArgoCD & CI/CD Failures (20 Questions)
- Docker & Container-Level Incidents (15 Questions)
- General Production Incident Handling & Postmortems (7 Questions)

Grand Total: 406 unique questions (254 concept + 152 scenario)

PART A

Concept-Level Production Questions

Architecture & High Availability

1. Jenkins master crash ho jaaye production me — recovery strategy kya hogi (backup, JENKINS_HOME restore)?
2. Jenkins ko High Availability (HA) me kaise setup karte ho (active-passive vs active-active)?
3. Master-Agent (Slave) architecture ka production use case kya hai aur agent scaling kaise karoge?
4. Jenkins ko Kubernetes pod agents ke saath integrate kaise karte ho (Kubernetes plugin, dynamic agents)?
5. JENKINS_HOME ka backup automation production me kaise design karte ho (thinBackup, S3 sync)?
6. Jenkins ka disaster recovery plan (config-as-code — JCasC) kaise banate ho?
7. Jenkins upgrade production environment me zero-downtime ke saath kaise karte ho?
8. Jenkins controller ko horizontally scale kaise karte ho jab load bahut zyada ho jaaye?
9. Static agents vs dynamic (ephemeral) agents — production me trade-offs kya hain?
10. Jenkins ko multi-region/DR setup ke saath kaise design karte ho?

Pipeline Design

11. Declarative vs Scripted pipeline — production me kab kaunsa prefer karoge aur kyun?
12. Shared Libraries kya hote hain aur multi-team environment me inka use kaise karte ho?
13. Parallel stages aur matrix builds ka production use case kya hai?
14. Pipeline me approval gates (manual input step) production release ke liye kaise design karte ho?
15. Multibranch pipeline production repos ke saath kaise scale karte ho?
16. Pipeline failure ke baad rollback strategy kaise design karoge?
17. Blue-Green ya Canary deployment Jenkins pipeline se kaise implement karoge?
18. Post-build actions (post { success/failure/always }) production notification ke liye kaise structure karte ho?
19. Pipeline ko modular kaise banate ho (reusable stages, shared library functions)?
20. Environment-specific pipeline logic (dev/UAT/prod) ko ek hi Jenkinsfile me kaise manage karte ho?

Security

21. Jenkinsfile me credentials securely kaise manage karte ho (Credentials Binding, Vault integration)?
22. Jenkins security hardening ke steps kya hain (RBAC, Matrix auth, CSRF protection)?
23. Secrets rotation ho jaaye to running/scheduled pipelines par impact kaise minimize karoge?
24. Jenkins daemon/controller ko network level pe kaise secure karte ho (reverse proxy, TLS)?
25. Role-Based Access Control (RBAC) multi-team production Jenkins me kaise design karte ho?
26. Plugin vulnerabilities production me kaise track aur patch karte ho?
27. Script Security plugin aur Groovy sandbox ka production role kya hai?
28. API tokens vs passwords — production automation ke liye kaunsa best practice hai?

- 29. Audit logging (who ran what build) production compliance ke liye kaise enable karte ho?
- 30. Agent-to-master communication securely (SSH/JNLP over TLS) kaise configure karte ho?

CI/CD & Integration

- 31. Jenkins job trigger karne ke different tareeke (webhook, poll SCM, cron) — production best practice kaunsa?
- 32. GitHub/GitLab webhook fail ho jaaye production me to troubleshooting steps kya honge?
- 33. Jenkins ko SonarQube/Trivy jaise quality-security tools ke saath pipeline me kaise integrate karte ho?
- 34. Build artifacts ko securely store/retrieve karne ka production approach (Nexus/Artifactory/S3)?
- 35. Docker image build-push Jenkins pipeline se production registry (ECR) tak kaise automate karte ho?
- 36. Jenkins se Kubernetes/Helm deployment automate karne ka production workflow kya hoga?
- 37. GitOps (ArgoCD) ke saath Jenkins CI ka integration kaise design karte ho?
- 38. Jenkins pipeline me automated rollback trigger (health check failure pe) kaise implement karte ho?
- 39. Multiple environments (UAT → Staging → Prod) ke beech promotion pipeline kaise design karte ho?
- 40. Notification/alerting (Slack, email, PagerDuty) production incident response me kaise integrate karte ho?

Performance, Scaling & Troubleshooting

- 41. Jenkins performance tuning kaise karte ho (heap size/JVM tuning, executor count, plugin bloat)?
- 42. Build queue bahut lambi ho jaaye production me to root cause aur solution kya hoga?
- 43. Jenkins controller high CPU/memory usage ka troubleshooting approach kya hoga?
- 44. Flaky tests/builds ko production pipeline me kaise handle karte ho (retry logic, quarantine)?
- 45. Jenkins log rotation aur disk space management production me kaise karte ho?
- 46. Build timeout aur hung jobs ko production me kaise detect aur handle karte ho?
- 47. Concurrent builds ke beech resource conflict (same environment) kaise avoid karte ho (lockable resources)?
- 48. Plugin compatibility issue ke baad production Jenkins break ho jaaye to rollback strategy kya hogi?
- 49. Jenkins metrics (build time, success rate, queue time) monitoring ke liye kaise expose karte ho (Prometheus plugin)?
- 50. Ek critical production release ke dauraan Jenkins hi down ho jaaye — contingency plan kya hoga?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Terraform

39 Questions (concept-level; state-disaster & destroy-recovery scenarios moved to Part B)

State Management

1. Remote state ko dusre Terraform project me kaise consume karte ho (terraform_remote_state)?

Safety & Guardrails

2. terraform plan me unexpected destroy dikhe to root cause kaise identify karoge?
3. Drift detection — manual changes ho jaayein to Terraform kaise handle karta hai?
4. Terraform apply production me team collaboration ke saath safe kaise banate ho (state locking, approval gate)?
5. Custom validation (precondition/postcondition blocks) production reliability ke liye kaise use karte ho?
6. lifecycle { ignore_changes } ka production use case kya hai aur kab risky ho sakta hai?
7. Policy as Code (Sentinel/OPA/Checkov) Terraform governance ke liye production me kaise implement karte ho?
8. IAM permission boundary/SCP se Terraform ko destructive actions se kaise restrict karte ho?
9. Force-unlock state lock production me kab aur kaise safely karte ho?

Module & Code Design

10. Terraform module design production-grade reusable infra ke liye kaise karte ho?
11. Module versioning aur registry (private module registry) production me kaise manage karte ho?
12. DRY principle Terraform code me kaise maintain karte ho large infra ke liye?
13. Root module vs child modules ka structure production repo me kaise design karte ho?
14. Terraform code review process (PR-based plan output check) production ke liye kaise design karte ho?
15. count vs for_each — production me kab kaunsa use karoge aur kyu?
16. Dynamic blocks ka production use case kya hai?
17. Terraform variables validation (variable block ke andar validation) production me kaise use karte ho?
18. Environment-specific configuration (.tfvars files) production me kaise organize karte ho?
19. Terraform code me naming convention aur tagging strategy production governance ke liye kya honi chahiye?

CI/CD & Automation

20. Terraform ke saath CI/CD pipeline (plan on PR, apply on merge) production ke liye kaise design karte ho?
21. Terraform Cloud/Enterprise vs self-managed backend — production choice ka reasoning?
22. Automated terraform plan output ko human-readable review ke liye pipeline me kaise present karte ho?
23. Multiple environments ke deployment (dev→stage→prod) Terraform pipeline se kaise promote karte ho?

24. Terraform apply pipeline me approval gate (manual intervention) kaise implement karte ho?
25. Terraform provider version pinning aur upgrade strategy production me kaise handle karte ho?
26. Terraform pipeline me secrets injection (CI/CD variables se) kaise securely karte ho?
27. Automated testing Terraform code ke liye (Terratest, terraform validate) production pipeline me kaise integrate karte ho?
28. Rollback strategy Terraform-managed infra ke liye kya hoti hai (kyunki traditional rollback nahi hota)?

Scaling, Performance & Real-World Scenarios

29. Large-scale infra (100+ resources) me Terraform performance/plan time kaise optimize karte ho?
30. Cross-account/cross-region resource provisioning Terraform se kaise manage karte ho?
31. Terraform ke saath zero-downtime infra changes (e.g., ASG/Launch Template replace) kaise achieve karte ho?
32. Terraform me provider aliasing multi-region/multi-account setup ke liye kaise use karte ho?
33. Data sources (data block) production me kab use karte ho vs resource create karna?
34. Terraform ke saath Kubernetes resources (via kubernetes/helm provider) production me manage karna kab appropriate hai?
35. Terraform state me hazaaron resources ho jaayein to team collaboration aur ownership kaise split karte ho?
36. Terraform ka use karke immutable infrastructure pattern production me kaise implement karte ho?
37. Ek production outage hua ho Terraform apply ki wajah se — post-incident process (root cause, prevention) kya hoga?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Docker

44 Questions (concept-level; OOM/daemon-crash/disk-full/secrets-leak scenarios moved to Part B)

Image Building & Optimization

1. Docker image size optimize kaise karte ho (multi-stage builds, distroless/alpine images)?
2. Docker layer caching build pipeline speed ke liye kaise optimize karte ho?
3. .dockerignore production build me kyu zaroori hai aur na hone se kya problem aati hai?
4. Multi-arch image builds (ARM/AMD) production ke liye kaise banate ho (buildx)?
5. Image tagging strategy (immutable tags, semantic versioning, latest avoid karna) production releases ke liye kya honi chahiye?
6. Base image selection production security/size ke liye kaise karte ho?
7. Docker build cache invalidation issues (dependency layer change) kaise handle karte ho?
8. Dockerfile best practices (layer order, single RUN combine) production efficiency ke liye kya hain?
9. Build args vs environment variables — production image build me kab kaunsa use karoge?

Security

10. Container security hardening (non-root user, read-only filesystem) kaise karte ho?
11. Docker daemon ki configuration production security ke liye kaise harden karte ho (TLS, socket permissions)?
12. Seccomp/AppArmor profiles production containers me kaise apply karte ho?
13. Container escape vulnerability se production host ko kaise protect karte ho?
14. Docker registry (private ECR/Harbor) security aur image signing (Cosign/Notary) kaise implement karte ho?
15. Privileged containers production me kab avoid karte ho aur kyu risky hote hain?
16. Container-to-container secrets sharing production me securely kaise karte ho?
17. Docker socket (/var/run/docker.sock) expose karna production me kyu risky hai?
18. Supply chain security (base image provenance, SBOM generation) production me kaise ensure karte ho?

Networking

19. Docker networking modes (bridge, host, overlay, none) production use case ke hisaab se kab choose karoge?
20. Overlay network multi-host container communication (Swarm) me kaise kaam karta hai?
21. Container DNS resolution issues production me kaise troubleshoot karte ho?
22. Port mapping conflicts production multi-container environment me kaise avoid karte ho?
23. Docker network isolation (multiple networks per container) production security ke liye kaise design karte ho?

Storage & Data

24. Docker volumes vs bind mounts — production data persistence ke liye kab kaunsa use karoge?

- 25. Docker storage drivers (overlay2) production performance ko kaise affect karte hain?
- 26. Container data backup/restore strategy production me kaise design karte ho?
- 27. Ephemeral vs persistent storage containers me production workload ke hisaab se kaise decide karte ho?
- 28. Volume permission issues (UID/GID mismatch) production me kaise troubleshoot karte ho?

Runtime, Resource Management & Troubleshooting

- 29. Container resource limits (CPU/memory cgroups) production me kaise set aur tune karte ho?
- 30. Zombie processes/PID 1 issue container me kaise handle karte ho (init process, --init flag)?
- 31. Container restart policies production reliability ke liye kaise configure karte ho?
- 32. Container logs production me centralized logging (ELK/EFK/Fluentd) ke saath kaise manage karte ho?
- 33. High CPU/memory consuming container production me kaise identify karte ho (docker stats, cAdvisor)?
- 34. Container health checks (HEALTHCHECK instruction) production me kaise design karte ho?
- 35. Container startup slow ho to root cause kaise identify karte ho (image size, entrypoint script)?

Orchestration, CI/CD & Real-World Scenarios

- 36. Docker Compose vs Kubernetes — production scaling ke liye limitation kya hai Compose ki?
- 37. Docker Swarm vs Kubernetes — production orchestration ke liye trade-offs kya hain?
- 38. CI/CD pipeline me Docker image build-push (to ECR/Harbor) automate kaise karte ho?
- 39. Container-based blue-green/canary deployment production me kaise implement karte ho?
- 40. Docker Compose production environment me use karna kab appropriate hai aur kab nahi?
- 41. Multi-container application (app+db+cache) production me Compose/Swarm se kaise orchestrate karte ho?
- 42. Docker daemon upgrade production host par zero-downtime ke saath kaise karte ho?
- 43. Container image rollback strategy production incident ke time kaise implement karte ho?
- 44. Rootless Docker production me kab use karte ho aur uske limitations kya hain?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Kubernetes

44 Questions (concept-level; CrashLoopBackOff/etcd/DNS/Velero/NotReady/Secrets-choice scenarios moved to Part B)

Cluster Architecture & Troubleshooting

1. Control plane component (API server/scheduler/controller-manager) down ho jaaye to impact aur recovery kya hoga?
2. Cluster upgrade (control plane + worker nodes) zero downtime ke saath kaise karte ho?
3. Pod Pending state me stuck rahe to troubleshooting steps kya honge (resource, scheduling constraints)?
4. Node resource pressure (DiskPressure/MemoryPressure) production me kaise handle karte ho?
5. kubectl describe/kubectl logs/kubectl events ka use production debugging me kaise karte ho step-by-step?
6. Kubelet crash ho jaaye node pe to us node ke pods ka kya hota hai?

Workload Management

7. Rolling update vs Recreate vs Blue-Green vs Canary deployment strategy — trade-offs kya hain?
8. DaemonSet vs Deployment vs StatefulSet — production use case differences?
9. Resource requests/limits sahi tarike se kaise set karte ho (OOMKilled issue avoid karne ke liye)?
10. Liveness, Readiness, Startup probes — misconfiguration se production outage kaise hota hai?
11. Init containers ka production use case kya hai?
12. Job/CronJob production me failed runs ko kaise handle/retry karte ho?
13. Pod affinity/anti-affinity aur node affinity production scheduling ke liye kaise use karte ho?
14. Taints and Tolerations production me dedicated nodes ke liye kaise use karte ho?
15. ConfigMap/Secret update hone par running pods ko kaise reflect karwate ho (restart trigger)?
16. Sidecar container pattern production me kab aur kyu use karte ho?

Scaling & Autoscaling

17. Horizontal Pod Autoscaler (HPA) vs Vertical Pod Autoscaler (VPA) — production me kab use karoge?
18. Cluster Autoscaler ka node scaling behavior aur cost optimization se relation kya hai?
19. Custom metrics (Prometheus Adapter) ke basis pe HPA production me kaise configure karte ho?
20. Pod Disruption Budget (PDB) ka production use case kya hai?
21. Multi-dimensional scaling (HPA + Cluster Autoscaler together) production me kaise coordinate karte ho?

Networking

22. Pod-to-Pod communication fail ho rahi ho to Network Policy/CNI troubleshooting kaise karoge?
23. Service types (ClusterIP, NodePort, LoadBalancer, ExternalName) — production me kab kaunsa use karoge?
24. Ingress Controller high traffic me bottleneck ban jaaye to scaling/troubleshooting approach?

- 25. Network Policies production multi-tenant cluster me kaise design karte ho?
- 26. Service Mesh (Istio/Linkerd) production me kab implement karna chahiye aur kyun?

Storage

- 27. Persistent storage (PV/PVC) production me StatefulSets ke saath kaise manage karte ho?
- 28. StorageClass aur dynamic provisioning production me kaise design karte ho?
- 29. PVC stuck Pending state me ho to troubleshooting steps kya honge?
- 30. ReadWriteOnce vs ReadWriteMany volumes — production workload ke hisaab se kab kaunsa use karoge?

Security & Multi-Tenancy

- 31. RBAC design kaise karte ho multi-team production cluster ke liye?
- 32. Namespace isolation aur ResourceQuota kaise design karte ho multi-tenant cluster me?
- 33. Pod Security Standards/Admission Controllers production me kaise enforce karte ho?
- 34. Service Account permissions least-privilege ke hisaab se production me kaise design karte ho?

Observability, Upgrades & Real-World Scenarios

- 35. Kubernetes me observability stack (Prometheus, Grafana, logging) kaise design karte ho?
- 36. Cluster me sudden high memory/CPU usage ho jaaye to root cause kaise identify karte ho?
- 37. Application deployment ke baad 5xx errors spike ho jaayein — troubleshooting approach kya hoga?
- 38. GitOps (ArgoCD/Flux) ke saath production deployment pipeline kaise design karte ho?
- 39. Multi-cluster setup (DR/region failover) production me kaise design karte ho?
- 40. Node drain/cordon production maintenance ke waqt kaise safely karte ho?
- 41. Helm charts production release management ke liye kaise use karte ho (rollback, versioning)?
- 42. Kubernetes API deprecation (API version change) production upgrade ke waqt kaise handle karte ho?
- 43. Ek real production incident — deployment ke baad pods sab crash ho gaye — post-incident RCA process kya hoga?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Cluster Setup & Architecture

1. EKS cluster upgrade production me kaise safely karte ho (control plane + node groups)?
2. Managed Node Groups vs Self-managed nodes vs Fargate — production me kaunsa kab choose karoge?
3. EKS me private cluster (no public endpoint) setup aur access strategy kya hogi?
4. Multi-AZ high availability EKS cluster kaise design karte ho?
5. EKS control plane vs data plane — production troubleshooting me responsibility split kaise samajhte ho?
6. EKS Fargate profiles ka production limitation kya hai aur kab avoid karoge?
7. Multi-cluster EKS setup (DR/region failover) kaise design karte ho?
8. EKS Anywhere/on-prem use case production me kab relevant hota hai?
9. EKS cluster ka version deprecation handle karna production me kaise karte ho?
10. New EKS cluster onboarding process production me end-to-end kaise design karte ho?

Networking

11. EKS networking — VPC CNI limitations (IP exhaustion) production me kaise solve karte ho?
12. Load balancing — ALB Ingress Controller vs NLB — production traffic ke liye kaunsa use karoge?
13. EKS me security groups for pods (SGP) production me kab use karte ho?
14. Cross-VPC/cross-account networking EKS cluster ke liye kaise design karte ho?
15. Custom CNI (Calico/Cilium) EKS me kab aur kyu use karte ho?
16. EKS me DNS resolution issues (CoreDNS) production me kaise troubleshoot karte ho?
17. Service mesh (App Mesh/Istio) EKS production me kab implement karte ho?
18. EKS me private endpoint access aur VPC peering/Transit Gateway integration kaise karte ho?
19. Pod networking troubleshooting (ENI limits hit ho jaayein) production me kaise karte ho?
20. Ingress + WAF integration EKS production me kaise design karte ho?

Security & IAM

21. EKS me IAM Roles for Service Accounts (IRSA) kaise configure aur secure karte ho?
22. EKS me Secrets encryption (KMS envelope encryption) kaise implement karte ho?
23. EKS cluster me Pod Security Standards/Admission Controllers kaise enforce karte ho?
24. Cross-account/cross-cluster EKS access kaise manage karte ho?
25. aws-auth ConfigMap production me kaise manage karte ho (RBAC mapping)?
26. EKS cluster ka complete audit/compliance (CIS Benchmark) kaise implement karte ho?
27. Node IAM role permissions least-privilege ke hisaab se production me kaise design karte ho?
28. Image scanning aur admission control (OPA Gatekeeper/Kyverno) EKS production me kaise implement karte ho?

- 29. EKS API server access production me network-level (private endpoint + bastion) kaise secure karte ho?
- 30. Secrets rotation EKS workloads ke liye (Vault/External Secrets Operator) kaise automate karte ho?

Scaling & Cost Optimization

- 31. Cluster Autoscaler vs Karpenter — production scaling ke liye kaunsa better hai aur kyun?
- 32. Cost optimization EKS me Spot Instances ke saath production workloads ke liye kaise karte ho?
- 33. EKS node group ka right-sizing (instance type selection) production me kaise decide karte ho?
- 34. HPA/VPA ke saath Cluster Autoscaler/Karpenter ka coordination production me kaise kaam karta hai?
- 35. EKS me node draining/termination handling (Spot interruption) kaise handle karte ho?
- 36. Service quotas/limits (ENI, IP, node count) EKS me hit ho jaayein to troubleshooting kaise karoge?
- 37. Reserved capacity/Savings Plans EKS node groups ke liye production cost optimization me kaise use karte ho?
- 38. Bin packing/resource utilization improve karne ke liye production me kya strategy follow karte ho?

Observability, Add-ons & Operations

- 39. EKS logging aur monitoring (Control Plane logs, CloudWatch Container Insights) kaise enable karte ho?
- 40. Add-ons (VPC CNI, CoreDNS, kube-proxy) upgrade karte waqt production risk kaise minimize karoge?
- 41. Prometheus/Grafana EKS cluster ke saath production observability ke liye kaise integrate karte ho?
- 42. EKS cluster health check aur alerting strategy production me kaise design karte ho?
- 43. EKS control plane logs (audit, authenticator) troubleshooting me kaise use karte ho?
- 44. Node-level troubleshooting (kubelet issues, disk pressure) EKS worker nodes pe kaise karte ho?

CI/CD & Real-World Scenarios

- 45. GitOps (ArgoCD/Flux) ke saath EKS production deployment pipeline kaise design karte ho?
- 46. Terraform/eksctl — production EKS provisioning ke liye kaunsa approach best hai aur kyu?
- 47. Blue-Green cluster upgrade strategy (naya cluster bana ke migrate karna) production me kab use karte ho?
- 48. CI/CD pipeline se EKS tak deployment (Helm/Kustomize) kaise automate karte ho?
- 49. Production incident — EKS nodes suddenly NotReady ho gaye — troubleshooting approach kya hoga?
- 50. Ek real scenario — EKS cluster me pods scheduling stop ho gaya (resource/IP exhaustion) — RCA aur fix kya hoga?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Architecture & Design

1. Ansible ka agentless architecture production me kaise kaam karta hai (SSH-based execution)?
2. Ansible Playbook structure (roles, tasks, handlers, vars) production-grade project me kaise organize karte ho?
3. Ansible roles ka reusable design production me kaise banate ho (multiple projects ke liye)?
4. Static inventory vs dynamic inventory — production me (especially cloud/AWS) kab kaunsa use karoge?
5. Ansible Vault production me sensitive data (passwords, keys) encrypt karne ke liye kaise use karte ho?
6. Ansible Tower/AWX production me kab use karte ho aur plain Ansible se kya benefit milta hai?
7. Group_vars aur host_vars ka production use case multi-environment (dev/stage/prod) ke liye kya hai?
8. Ansible Galaxy roles production me use karte waqt security/versioning kaise handle karte ho?

Idempotency & Execution

9. Idempotency Ansible playbooks me kaise ensure karte ho (especially custom shell/command modules ke saath)?
10. handlers aur notify production me kab use karte ho (e.g., service restart on config change)?
11. check mode (--check) aur --diff production me changes verify karne ke liye kaise use karte ho?
12. Playbook execution beech me fail ho jaaye to production recovery/resume strategy (--start-at-task) kya hogi?
13. Tags production me selective task execution ke liye kaise use karte ho?
14. serial keyword rolling deployment (batch-wise execution) production me kaise use karte ho?
15. changed_when/failed_when custom conditions production me kab define karte ho?

Security & Access

16. SSH key management production me multiple servers ke liye kaise secure karte ho?
17. Ansible Vault password rotation production me kaise handle karte ho bina downtime ke?
18. become/become_user (privilege escalation) production me securely kaise configure karte ho?
19. Sensitive output (secrets) logs me leak na ho, iske liye no_log production me kaise use karte ho?
20. Ansible control node ki security (jahan se playbooks run hote hain) production me kaise harden karte ho?

CI/CD, Scaling & Real-World Scenarios

21. Ansible ko Jenkins/GitLab CI pipeline ke saath production automation ke liye kaise integrate karte ho?
22. Large-scale infra (hundreds of servers) me Ansible execution performance kaise optimize karte ho (forks, mitogen)?
23. Ansible ke through zero-downtime rolling deployment production me kaise achieve karte ho?

24. Configuration drift (manual server changes) production me Ansible se kaise detect/fix karte ho?
25. Ansible aur Terraform ka combined workflow production me kaise design karte ho (provisioning + config management)?
26. Multi-OS (RHEL, Ubuntu, Amazon Linux) environment me ek hi playbook production me kaise manage karte ho?
27. Playbook testing (Molecule) production release ke pehle kaise implement karte ho?
28. Error handling (block/rescue/always) production playbooks me kaise design karte ho?
29. Ansible se patch management/security updates production servers pe automated kaise karte ho?
30. Ek production incident — playbook run ke baad service down ho gayi — rollback aur RCA approach kya hoga?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

PART B

Real Disaster / Incident Scenario Questions ("kisi ne galti se delete kar diya" style)

Kubernetes — Cluster & Workload Failures

30 Questions

1. Someone accidentally ran `kubectl delete namespace production`. What is your immediate recovery plan?
2. How do you recover a deleted Deployment if you don't have the YAML manifest saved anywhere?
3. A fresher accidentally deleted the entire EKS/GKE cluster. What are your first 3 steps?
4. How do you take and restore etcd backups in a self-managed Kubernetes cluster?
5. Your production pods are stuck in `CrashLoopBackOff` at 3 AM. Walk through your debugging steps.
6. A `kubectl apply` with a wrong context accidentally applied a dev manifest to prod. How do you detect and roll back?
7. How would you recover if `PersistentVolumeClaims` got deleted along with their `PersistentVolumes` (reclaim policy = Delete)?
8. What's the difference between Retain, Delete, and Recycle reclaim policies, and why does this matter during disasters?
9. Your Ingress controller pod crashed and no traffic is reaching any service. How do you triage?
10. How do you handle a scenario where a `ConfigMap` was updated with bad values and it crashed all dependent pods?
11. A node in your cluster went `NotReady` and workloads aren't rescheduling. What do you check?
12. How would you recover a cluster where the control plane is down but worker nodes are still running workloads?
13. Someone force-deleted a `StatefulSet` along with its PVCs holding a production database. What now?
14. How do you design `PodDisruptionBudgets` to prevent an accidental `kubectl drain` from taking down all replicas?
15. A rogue HPA (Horizontal Pod Autoscaler) scaled a deployment to 0 replicas in production. How do you find out why and fix it?
16. How do you recover from a botched rolling update where the new pods are failing health checks but old pods are already terminated?
17. Explain how you'd restore a Kubernetes cluster from a Velero backup.
18. A namespace is stuck in `Terminating` state forever. How do you force-clean it safely?
19. How do you handle "resource quota exceeded" errors blocking new deployments during an incident?
20. Your cluster's DNS (CoreDNS) is failing intermittently. How do you diagnose and fix it?
21. How would you respond if someone deleted the `kube-system` namespace by mistake?
22. A malicious or buggy `CronJob` is spawning thousands of pods every minute. How do you stop the bleeding immediately?
23. How do you set up multi-region/multi-cluster failover so a single cluster deletion doesn't cause total downtime?
24. What is your RTO/RPO strategy for Kubernetes-hosted stateful applications?
25. How do you test disaster recovery for Kubernetes without impacting real production?
26. How would you recover if the container registry (ECR/GCR/Docker Hub) became unreachable during a scale-up event?

27. A node pool got accidentally deleted in GKE/EKS along with all its pods. What's your recovery sequence?

28. How do you prevent a single kubectl delete command from wiping out an entire environment (guardrails, RBAC, OPA/Gatekeeper)?

29. Explain how admission controllers (like Kyverno/OPA) can prevent destructive human errors in production.

30. How do you recover a corrupted etcd cluster with no recent snapshot?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Kubernetes — Secrets & Access Management

25 Questions

1. How do native Kubernetes Secrets differ from tools like HashiCorp Vault or Sealed Secrets in terms of security?
2. Someone committed a Kubernetes Secret YAML (base64 only) to a public GitHub repo. What's your incident response?
3. How would you rotate a leaked database credential used by 20 microservices in production without downtime?
4. How do you integrate External Secrets Operator with AWS Secrets Manager / Vault for automatic secret sync?
5. What's the risk of using ConfigMaps for storing sensitive data instead of Secrets?
6. How do you implement short-lived, auto-rotating credentials for pods accessing a cloud database?
7. Explain how Sealed Secrets works and why it's safer to store in Git than plain Secrets.
8. How would you audit who accessed a particular Secret in the last 30 days in Kubernetes?
9. How do you prevent a developer with `kubectl exec` access from reading production secrets from inside a pod?
10. How do you design RBAC so that a "read-only" role cannot still extract secret values via `kubectl get secret -o yaml`?
11. What is a Service Account token, and how would you respond if one got leaked?
12. How do you handle secret rotation for a Terraform-managed RDS instance without breaking currently running pods?
13. How would you implement Vault Agent Injector so secrets are never stored as Kubernetes objects at all?
14. How do you manage TLS certificate rotation automatically using cert-manager in production?
15. A pod is mounting an outdated Secret because it wasn't restarted after rotation. How do you detect and fix this pattern cluster-wide?
16. How do you prevent secrets from leaking into pod logs or crash dumps?
17. How would you set up namespace-level isolation so Team A's secrets are never visible to Team B?
18. Explain envelope encryption and how it applies to Kubernetes Secrets encryption at rest.
19. How do you handle secrets for CI/CD pipelines (GitHub Actions/GitLab CI) that deploy to Kubernetes?
20. What steps would you take if you discovered a Vault root token had been exposed in a Slack message?
21. How do you securely pass secrets between Terraform and Kubernetes without storing them in `.tfstate` in plaintext?
22. How do you rotate SSH keys used by CI runners that have cluster access, without breaking active deployments?
23. How would you design a "break-glass" emergency access procedure for production secrets?
24. How do you prevent secret sprawl across dozens of namespaces and multiple clusters?
25. What's your approach to secret management for a multi-tenant Kubernetes cluster shared across teams?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Terraform — State File & Infrastructure Disasters

30 Questions

1. Someone deleted the Terraform state file (terraform.tfstate) from the S3 backend. What is your recovery process?
2. How do you recover if terraform apply was run with a stale/local state file and it now conflicts with the real remote state?
3. Explain how Terraform state locking (via DynamoDB) prevents two engineers from corrupting state simultaneously.
4. A fresher ran terraform destroy on the wrong workspace and deleted production infrastructure. What's your response plan?
5. How do you recover resources that exist in the cloud but are missing from Terraform state (state drift)?
6. How would you import an existing manually-created AWS resource into Terraform state without recreating it?
7. What's your strategy for versioning and backing up Terraform state files?
8. How do you handle a scenario where terraform plan shows it wants to destroy and recreate a production database due to an immutable field change?
9. How would you recover if the S3 bucket storing your Terraform state was accidentally deleted?
10. Explain the danger of running terraform apply -auto-approve in a CI/CD pipeline without a plan review step.
11. How do you structure Terraform modules/workspaces to prevent someone from accidentally modifying prod while working on dev?
12. What happens if two team members run terraform apply at the same time without state locking, and how do you recover from the corruption?
13. How do you roll back an infrastructure change if the previous Terraform state version is unavailable?
14. Explain how you'd set up cross-region replication for your Terraform remote state for disaster recovery.
15. How do you handle "resource already exists" errors when Terraform state gets out of sync with real infrastructure?
16. Someone manually deleted a resource from the AWS Console that Terraform manages. What happens on the next apply and how do you fix it?
17. How would you safely refactor Terraform code (renaming resources/modules) without Terraform destroying and recreating them?
18. What is terraform state mv used for, and describe a real incident where you'd need it?
19. How do you prevent secrets (like DB passwords) from being stored in plaintext inside the .tfstate file?
20. How would you recover if your Terraform Cloud/Enterprise account got locked or deleted along with all workspace history?
21. Explain your process for testing a Terraform disaster-recovery runbook (like tabletop drills for infra teams).
22. How do you handle Terraform provider version mismatches breaking apply in production during an urgent hotfix?
23. What is a "tainted" resource in Terraform and how would you use it to recover from a partially-broken deployment?

24. How would you set up Terraform to require manual approval before any destroy action on tagged production resources?
25. How do you handle a scenario where Terraform partially applied a change and then crashed midway (partial apply)?
26. How would you design your Terraform pipeline so a single compromised CI token can't destroy your entire cloud account?
27. Explain blue-green infrastructure deployment using Terraform and how it helps during disaster recovery.
28. How do you back up and restore Terraform Cloud/Enterprise workspace configurations?
29. What's the difference between terraform state rm and actually destroying a resource, and when would each be dangerous?
30. How would you recover a Terraform-managed Kubernetes cluster if the underlying VPC was accidentally deleted?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Database Disaster Recovery

25 Questions

1. Your production database got dropped (DROP DATABASE) by mistake. What are your first actions in the next 5 minutes?
2. How do you design a Point-in-Time-Recovery (PITR) strategy for a production PostgreSQL/MySQL database?
3. Explain your backup retention policy and how you validate that backups are actually restorable (not just "taken").
4. How would you recover data if the last full backup is 20 hours old but you need data from 2 hours ago?
5. How do you handle a scenario where an application bug silently corrupted data over the last 6 hours before anyone noticed?
6. What is your strategy for testing database restore procedures regularly (backup verification drills)?
7. How do you set up cross-region database replication for disaster recovery in AWS RDS / Cloud SQL?
8. Explain the difference between logical backups (pg_dump) and physical/snapshot backups, and when each is appropriate.
9. How would you recover if a Kubernetes StatefulSet running your database lost its PVC due to node failure?
10. How do you prevent an engineer from accidentally running a destructive query directly on production?
11. What's your approach to database failover — how do you detect the primary is down and promote a replica automatically?
12. How would you handle a ransomware-style scenario where your database backups were also encrypted/corrupted?
13. Explain how you'd set up automated, encrypted, off-site backups that are immutable (can't be deleted even by an admin).
14. How do you recover from a schema migration that went wrong and locked/corrupted production tables?
15. What's your rollback plan if a database migration ran successfully but broke application compatibility?
16. How would you handle split-brain in a multi-master database replication setup after a network partition?
17. How do you calculate and justify RPO (Recovery Point Objective) and RTO (Recovery Time Objective) for a critical database to leadership?
18. What monitoring/alerting would you put in place to detect data corruption before it becomes a full-blown incident?
19. How do you handle secrets/credentials for database backup and restore automation scripts?
20. Explain your approach to database disaster recovery testing in a regulated environment (e.g., banking, healthcare).
21. How would you recover a managed database (RDS/Cloud SQL) if the entire cloud region went down?
22. What's your process for restoring a database to a new environment for debugging without affecting production?
23. How do you handle backup storage costs while still maintaining a robust long-term retention policy?

24. How would you detect and respond to a compromised database credential being used for data exfiltration?

25. What's your plan if the automated backup job silently failed for the past 2 weeks and nobody noticed?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

GitOps, ArgoCD & CI/CD Failures

20 Questions

1. Someone deleted a Kubernetes manifest file from your GitHub repo, and ArgoCD auto-synced and deleted the live resource in production. How do you recover?
2. How does ArgoCD's "self-heal" and "prune" feature increase risk during accidental Git operations, and how do you mitigate it?
3. How would you roll back a bad deployment using ArgoCD's application history?
4. Explain your strategy for protecting the main/master branch that ArgoCD watches, to prevent unreviewed changes from auto-deploying.
5. How do you recover if the Git repository that ArgoCD syncs from becomes completely inaccessible or corrupted?
6. What happens if two different ArgoCD Applications point to overlapping resources, and how do you debug conflicts?
7. How would you set up ArgoCD to require manual sync approval for production but allow auto-sync for dev/staging?
8. How do you handle secrets in a GitOps workflow without ever committing them in plaintext to Git?
9. Explain how you'd recover a cluster's entire state using GitOps if the cluster itself was deleted (Git as source of truth).
10. How do you detect configuration drift between what's in Git and what's actually running in the cluster?
11. A CI/CD pipeline accidentally deployed an old, vulnerable image tag to production. How do you find out and fix it quickly?
12. How would you implement automated rollback in your CI/CD pipeline if post-deployment health checks fail?
13. How do you prevent a compromised CI/CD pipeline credential from being used to deploy malicious code to production?
14. Explain canary and blue-green deployment strategies and how they reduce blast radius during a bad release.
15. How would you handle a scenario where your image registry got compromised and images were tampered with?
16. How do you ensure your CI/CD pipeline itself has disaster recovery (e.g., Jenkins/GitLab server going down)?
17. What's your process for auditing who triggered a specific production deployment and when?
18. How would you implement policy-as-code (OPA/Kyverno) to block deployments that don't meet security baselines?
19. How do you handle a "config drift" incident where manual kubectl edit changes were overwritten by the next GitOps sync?
20. What's your incident response plan if a supply-chain attack injected malicious code into a widely-used base Docker image you depend on?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

Docker & Container-Level Incidents

15 Questions

1. A container is consuming 100% CPU/memory and taking down the entire host node. How do you diagnose and mitigate it?
2. How would you recover if the Docker daemon crashed on a production host running critical containers?
3. Explain how you'd debug a container that keeps restarting with exit code 137.
4. How do you prevent secrets from being baked into Docker image layers accidentally (and how would you clean up if it already happened)?
5. A Docker image was pushed with a hardcoded production database password. How do you remediate across all environments?
6. How would you recover disk space on a production node when Docker images/containers filled up the entire disk?
7. What's your strategy for scanning Docker images for vulnerabilities before they reach production?
8. How do you handle a scenario where a base image (e.g., node:18) got a critical CVE and needs to be patched across 50 services urgently?
9. Explain the difference between a container crash and a container being OOMKilled, and how you'd distinguish them in logs.
10. How would you troubleshoot a container that works locally but fails only in production (env parity issues)?
11. How do you prevent a docker system prune -a run by mistake from wiping out needed images on a production host?
12. What's your approach to signing and verifying container images to prevent supply-chain tampering?
13. How would you handle multi-architecture image builds breaking silently on ARM-based production nodes?
14. Explain how you'd set resource limits/requests properly to prevent one noisy container from starving others.
15. How do you recover logs from a container that crashed and was already removed by the orchestrator?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.

General Production Incident Handling & Postmortems

7 Questions

1. Walk me through your general incident response process from alert to resolution to postmortem.
2. How do you decide when to page on-call at 3 AM versus letting an issue wait until morning?
3. What does a blameless postmortem look like, and why does it matter for team culture?
4. How do you maintain a runbook culture so that any team member (not just you) can handle an outage?
5. How would you explain a major production outage and data-loss incident to non-technical leadership/stakeholders?
6. What's your approach to chaos engineering (e.g., using Chaos Mesh/Litmus) to proactively find these failure points before they happen in real incidents?
7. How do you balance “move fast” engineering culture with the guardrails needed to prevent these kinds of disasters?

Tip: Answer dete waqt concept ke saath ek real scenario/example (jo genuinely kiya ho) bhi add karo — production depth clearly dikhti hai.